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Vaishnav V. Rao

Education

2023-Present **University of Michigan, Ann Arbor, MI.**
- Candidate, Astronomy & Astrophysics PhD
  GPA: 4.0/4.0

2019–2023 **Indian Institute of Technology Bombay, Mumbai, India.**
- **Major:** Bachelor of Technology in Physics with Honours
  GPA: 9.51/10
- **Minor:** Mathematics
  GPA: 9.5/10

Research Publications

V. V. Rao et al. (2025). **Stars Born in the Wind: M82's Outflow and Halo Star Formation**. *The Astrophysical Journal.* 10.3847/1538-4357/adfdce.

V. V. Rao et al. (2023). **AGN Feedback Through Multiple Jet Cycles in the Seyfert Galaxy NGC 2639**. *Monthly Notices of the Royal Astronomy Society.* 524,1615 10.1093/mnras/stad1901.

Research Experience

May'24-Present **How do Starburst Outflows Affect Galaxy Evolution?.**
Guide: Prof. Eric Bell | Dept. of Astronomy, University of Michigan
- Analyzed **Hubble Space Telescope** resolved-star photometry of the stellar halo of the starburst galaxy M82 to characterize star formation triggered by a starburst outflow
- Used the color magnitude diagram (CMD) fitting software **MATCH** to deduce star formation histories and proposed novel mechanisms of outflow-induced star formation
- Analyzing **James Webb Space Telescope** photometry of M82 to understand the galaxy-wide star formation impact of starburst outflows

May'25-Present **How do Spiral Galaxies Evolve in a Merging Environment?.**
Guide: Prof. Eric Bell | Dept. of Astronomy, University of Michigan
- Analyzing and mapping the star formation history of the M81 galaxy disc to understand the evolution of spirals in a merging environment through archival **Hubble Space Telescope** resolved-star photometry.

Aug'23-May'24 **Search for Ultra Faint Dwarf Galaxies in the M81 Galaxy Group.**
Guide: Prof. Eric Bell | Dept. of Astronomy, University of Michigan
- Improved star-galaxy separation in **Subaru Hyper Suprime-Cam** photometry of the M81 group of galaxies using realistic morphological selections and **machine learning (ML) classifiers** to aid the search for ultra-faint dwarf **(UFD) satellites** in the stellar halo
- Searched for clustering of old, metal-poor red giant branch stars using **kernel density estimators** and ML-based clustering algorithms to identify candidate UFDs

May'22-Jul'22 **Episodic AGN Activity and Feedback in NGC 2639 .**
Guide: Dr. Preeti Kharb | National Centre for Radio Astrophysics(NCRA-TIFR), India
- Reduced the data of the **Giant Metrewave Radio Telescope** (GMRT) and the **Jansky Very Large Array** (JVLA) using the radio data reduction software CASA, to image the multiple radio jets of the Seyfert galaxy NGC 2639, originating from multiple minor mergers.
- Performed **synchrotron spectral ageing** of the radio lobes to estimate the timescale of minor mergers using the BRATS software and corroborated the results with theoretical estimates.
- Analysed the GALEX Near-UV and Far-UV images and CO molecular gas distribution data from the CARMA-EDGE survey to trace star formation and found clear signs of quenched star formation due to **AGN negative feedback** from the multiple jet episodes.

Aug'22- Jun'23 **Beyond Standard Model Explanations for Black Hole Mass Gap**.
Guide: Prof. Vikram Rentala | Dept. of Physics | Indian Institute of Technology, Bombay(IITB)
- Motivated by the detection of intermediate mass black holes by LIGO, worked on modifying the CO-reaction rate at astrophysical energies by incorporating a new **oxygen nucleus resonance** at low energies through the **R-Matrix** formalism to explain black holes within the classical mass gap.
- Investigated **dark matter capture** and **modified gravity** effects as viable mechanisms for increasing or decreasing final black hole mass.

Conferences

Finding Ultra Faint Dwarf Galaxies in M81's Stellar Halo, *Poster.*
Small Galaxies, Cosmic Questions-II, *Durham University, UK*
Dwarf Galaxies, Star Clusters, and Stellar Streams in the LSST Era, *University of Chicago, US*

Multiple Jets Driving AGN Feedback in the Seyfert NGC 2639?, *Poster.*
National Workshop on Galactic Inflows and Outflows on all Scales, *IUCAA, India*

Teaching and Mentorship

Fall 2024 & Winter 2024 **Graduate Student Instructor**, *Introduction to Astronomy & Astrophysics*.
Taught weekly labs to undergraduate astronomy majors to help them gain practical experience in spectroscopy, python, data analysis, and telescope observing amongst other astronomical techniques

Spring 2022 **Teaching Assistant**, *Quantum Physics & Applications*, IIT Bombay.
Handled a tutorial group of 42 undergraduates, presented problem solutions, revised concepts, solved doubts, conducted examinations, and graded exam papers.

May'21- Apr'22 **Department Academic Mentor**, *Student Mentorship Program*, IIT Bombay.
Assigned to 10 junior students to help them cope with any academic problems they may face in the Physics department or otherwise. Undertook initiatives beneficial to the department including course mentoring, organizing career sessions, and compiling academic records.

Outreach Activities

Jan'25- May'25 **Science Communications Fellowship**, *UM Museum of Natural History*.
Attended a workshop for science communication and participated in designing table-top activities for communicating astronomy research to the general public

Aug'24- Present **Graduate Outreach Coordinator**, *UM Dept. of Astronomy*.
Worked on reviving Ann Arbor **Astronomy on Tap**, a bar-hosted outreach event that delivers public lectures on astronomy and organizes astronomy-themed games and trivia.

Jun'20- May'21 **Convener, Kritika- The Astronomy Club**, *IIT Bombay*.
Organized institute-wide astronomy events with a 10-member team, coordinated the Krittika Summer Projects for UG–PhD participants, and led the Techfest Astrophysics Workshop for 200+ attendees on GRBs, gravitational waves, and EM counterparts.

Key Courses

Astronomy Cosmology, Galaxies, Stellar Structure & Evolution, High Energy Astrophysics, GW Astronomy

Physics Adv. General Relativity, Quantum Field Theory, Elementary Particle Physics, Nuclear Physics, Atomic & Molecular Physics, Data Analysis & Interpretation

Math Differential Geometry, Numerical Analysis, General Topology, Complex Analysis

Misc. Machine Learning, Scientific Writing, Optics & Spectroscopy Lab, Electronics Lab

Technical Skills

Languages Python, C++, Fortran, Git, HTML

Softwares MATCH, MESA, CASA, BRATS, Mathematica, DS9, Photoshop

Telescopes HST, Magellan (FourStar), Subaru HSC, GMRT